H∞ CONTROL

## What does the bound guarantee?

A stable closed-loop transfer T maps disturbance w to output z, with zero initial state.

Its H∞ norm is at most 2.

State the resulting L2-norm bound. Does it also imply $|z(t)| \le 2|w(t)|$ at every instant?

**TW-HINF-01** Graduate

Suggested: 1-2 minutes

H∞ CONTROL

## A small-gain certificate

A well-posed feedback loop connects stable blocks M and Δ.

You know $||M||_\infty = 2$ and $||\Delta||_\infty \le \delta$.

What range of δ gives a strict small-gain stability certificate? What can this test conclude at $\delta = 0.5$?

**TW-HINF-02** Graduate

Suggested: 1-2 minutes

LQR

## Which Riccati root?

For $dx/dt = x + u$, minimize $J = \int_0^\infty (x^2 + u^2)\, dt$.

The scalar Riccati equation is $2P - P^2 + 1 = 0$, with $u = -Px$.

Which root gives the stabilizing LQR solution? Give its closed-loop pole.

**TW-LQR-01** Graduate

Suggested: 1-2 minutes

LQR

## Scaling both cost weights

A standard unconstrained infinite-horizon LQR problem has a unique stabilizing gain K.

Replace Q by 10Q and R by 10R, leaving the dynamics unchanged.

Does K change? What happens to the optimal cost? Explain without solving a Riccati equation.

**TW-LQR-02** Graduate

Suggested: 1-2 minutes

LQG

## Poles after adding an estimator

A continuous-time LQG controller uses $u = -K\hat{x}$ and a Kalman estimator.

$A - BK$ has poles $-1$ and $-2$.
$A - LC$ has poles $-3$ and $-4$.

What are the four poles of the nominal combined system? Explain the separation argument.

**TW-LQG-01** Graduate
Suggested: 1-2 minutes

LQG

## A noisier sensor

In a standard unconstrained LQG design, keep A, B, C, the process noise and the LQR cost weights fixed.

Increase only the measurement-noise covariance.

Must the LQR state-feedback gain K change? Which design must be recomputed, and why?

**TW-LQG-02** Graduate
Suggested: 1-2 minutes

BANG-BANG CONTROL

## Accelerate, then brake

A unit-mass particle obeys $dx/dt = v$, $dv/dt = u$, $|u| \le 1$.

Move from $(x,v) = (0,0)$ to $(1,0)$ in minimum time.

Give the two bang values, the switching time and the total time. Briefly justify the times.

**TW-BB-01** Graduate
Suggested: 1-2 minutes

BANG-BANG CONTROL

## When the switching function vanishes

Use the MINIMIZATION convention for a Hamiltonian
$H = H_0 + \varphi(t)u$, with $|u| \le 1$.

Give the minimizing u when $\varphi > 0$ and when $\varphi < 0$.

If $\varphi$ is zero throughout an interval, does this condition determine u there? Name this type of arc.

**TW-BB-02** Graduate
Suggested: 1-2 minutes

NONLINEAR CONTROL

## Beyond a non-increasing V

Consider $dx/dt = y$ and $dy/dt = -x - y^3$.

Take $V = (x^2 + y^2)/2$.

Compute $dV/dt$. Can you conclude global asymptotic stability of the origin? Identify the invariant-set argument.

**TW-NL-01** Graduate

Suggested: 2 minutes

NONLINEAR CONTROL

## A local result has limits

For $dx/dt = -x + x^3$, the linearization at the origin is $dx/dt = -x$.

Is the origin globally asymptotically stable? Find its exact region of attraction on the real line and explain what limits it.

**TW-NL-02** Graduate

Suggested: 1-2 minutes

NONLINEAR CONTROL

## Exact feedback linearization

A system has full-state measurements and no input limits:
$dx_1/dt = x_2$
$dx_2/dt = \sin(x_1) + (1 + x_1^2)u$.

Choose $u(x_1,x_2)$ so that
$d^2x_1/dt^2 + 3\,dx_1/dt + 2x_1 = 0$.

Is your feedback singular anywhere?

**TW-NL-03** Graduate

Suggested: 1-2 minutes

NONLINEAR CONTROL

## Attraction to a cycle

In polar coordinates, a planar system obeys
$dr/dt = r(1 - r^2)$, $d\theta/dt = 1$.

For any initial radius $r > 0$, what motion attracts the trajectory?

If two trajectories start on that motion with different phases, do their positions converge to each other?

**TW-NL-04** Graduate

Suggested: 1-2 minutes